



HARUTO NISHIMURA

Machine Learning Engineer

Middle · Osaka, Japan

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PROFILE

Machine Learning Engineer with a background in embedded systems and industrial automation. Builds computer-vision solutions for factory inspection, from dataset design and model evaluation through optimized edge deployment. Comfortable working across Python, C++ and Rust, with practical experience balancing accuracy, latency, memory use and maintainability on production devices.

CORE SKILLS

Computer Vision · Edge Inference · Model Evaluation · Python · C++ · Rust · PyTorch · OpenCV · ONNX Runtime · Embedded Linux · Data Annotation & Dataset Design

EXPERIENCE

Kansai Vision Works | Machine Learning Engineer

05/2023 - Present

- Developed defect-detection and surface-segmentation models for metal parts, establishing repeatable validation splits by production line and material batch.
- Converted PyTorch models to ONNX Runtime and TensorRT pipelines on NVIDIA edge devices; reduced median inspection latency from 180 ms to 72 ms.
- Created an evaluation harness covering precision, recall, false rejects and drift across weekly factory samples, helping operations identify lighting-related degradation before release.

Tech: Python, PyTorch, OpenCV, ONNX Runtime, TensorRT, C++

Asteron Robotics | Embedded Software Engineer

04/2020 - 04/2023

- Implemented C++ firmware and Linux services for camera-equipped inspection stations, including sensor synchronization, buffering and health monitoring.
- Introduced reproducible camera-calibration and image-capture tools that shortened setup work for a new production line from several days to one day.
- Partnered with controls and manufacturing engineers to collect labelled samples and define acceptance thresholds for an initial visual-inspection prototype.

Tech: C++, Rust, Embedded Linux, Qt, OpenCV, GitLab CI

Osaka Mechatronics Lab | Embedded Systems Intern

04/2019 - 03/2020

- Built a Rust-based telemetry collector for ARM evaluation boards and documented timing, memory and failure-recovery behaviour.
- Assisted with C firmware tests and camera-interface debugging for a prototype conveyor monitoring system.

Tech: Rust, C, ARM, Linux, Python

EDUCATION

Osaka Institute of Technology

BEng, Computer Engineering · 2015-2019

LANGUAGES

Japanese: native · English: B2